

Titanic Dataset — Exploratory Data Analysis

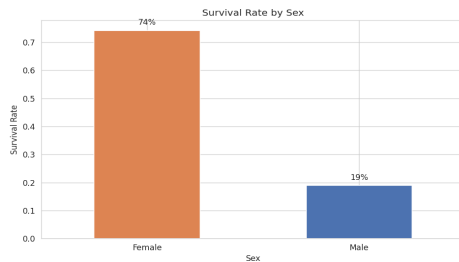
ZYNXIS AI/ML Internship · Month 1, Week 1 Deliverable · 891 passenger records

Dataset & Cleaning Approach

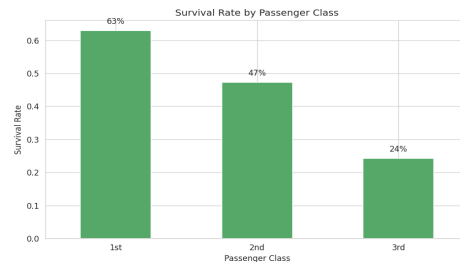
The Titanic dataset contains 891 passenger records with 12 attributes (class, sex, age, fare, family relations, cabin, port of embarkation, and survival outcome). Three columns had missing data: **Age** (177 missing, 19.9%), **Cabin** (687 missing, 77.1%), and **Embarked** (2 missing, 0.2%).

- **Age** — imputed using the median age within each (Passenger Class, Sex) group, since age varies meaningfully across those groups rather than using one global median.
- **Embarked** — only 2 missing values, filled with the mode (most common port).
- **Cabin** — too sparse (77% missing) to impute reliably, so converted into a binary feature *HasCabin* instead of dropping it, preserving its signal as a proxy for class/fare.
- **Dropped** PassengerId, Name, and Ticket as non-analytical identifiers.

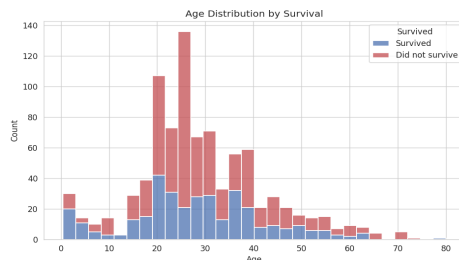
Key Visual Findings



Survival rate by sex: 74% of women survived vs. only 19% of men — the single strongest signal in the data.



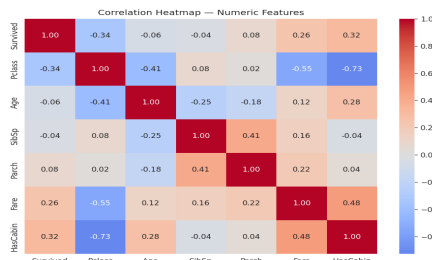
Survival rate by class: 1st class passengers survived far more often than 2nd or 3rd class.



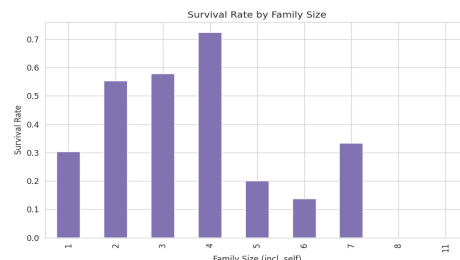
Age distribution by survival: young children show relatively better survival odds.



Fare by class: a clear, expected gradient — 1st class fares are dramatically higher and more variable.



Correlation heatmap: Fare and Pclass are the strongest numeric correlates of Survived.



Survival by family size: solo travelers and very large families fared worse than small families (2-4).

Conclusions

- Sex was the dominant predictor of survival, consistent with a 'women and children first' evacuation priority.
- Passenger class (a proxy for cabin location and fare) was the second strongest factor — wealthier, higher-deck passengers survived more often.
- Family size had a non-linear effect: small families (2-4 people) survived better than solo travelers or very large families.
- Cabin-recording status (HasCabin) correlates with survival mainly because it correlates with class, not because of cabin location itself.